

Rational and Irrational Numbers

Grade 8 / Pre-Algebra

Focus: perfect-square roots, non-perfect roots, decimals and fractions, comparing and ordering

The test you are using. A number is **rational** when it can be written as $\frac{p}{q}$ with p and q whole numbers and $q \neq 0$. Every terminating decimal and every repeating decimal passes that test. A number is **irrational** when no such fraction exists — its decimal never ends and never repeats.

Model to work from. Keep this table of perfect squares beside you. A square root is exact — and therefore rational — exactly when the number under the radical appears in the bottom row.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196

For every problem, state **rational** or **irrational** and give the reason in the work space. Write the value the problem asks for on the answer line.

1. Is $\sqrt{49}$ rational or irrational? Support your answer by writing $\sqrt{49}$ as a ratio of two whole numbers, then give its value.

Answer: _____

2. Look up $\sqrt{196}$ in the table above. Is it rational or irrational, and what is its exact value?

Answer: _____

3. Complete this row of the conversion table for the decimal 0.35, then classify it.

Decimal	Fraction as read aloud	Lowest terms	Rational?
0.35			

Write the lowest-terms fraction on the answer line.

Answer: _____

4. The number 50 is not in the table of perfect squares. (a) Between which two consecutive whole numbers does $\sqrt{50}$ lie? (b) Is $\sqrt{50}$ rational or irrational? (c) Give $\sqrt{50}$ rounded to two decimal places, and explain why that rounded value is not the same number as $\sqrt{50}$. Write the rounded value on the answer line.

Answer: _____

- 5.** Order these three numbers from least to greatest, and mark the one that is irrational:

$$\sqrt{13}, \quad 3.5, \quad \frac{18}{5}$$

Show how you compared them — rewriting each as a decimal is a good start.

Answer: _____

- 6.** The decimal $0.\overline{45}$ repeats forever. **(a)** Explain why a decimal that repeats forever can still be rational. **(b)** It is known that $0.\overline{45} = \frac{45}{99}$. Write that fraction in lowest terms on the answer line.

Answer: _____

- 7.** **(a)** Between which two consecutive whole numbers does $\sqrt{30}$ lie? Justify it with two entries from the square table. **(b)** Is $\sqrt{30}$ rational or irrational? **(c)** Give $\sqrt{30}$ rounded to two decimal places on the answer line.

Answer: _____

- $$\sqrt{2} \underline{\hspace{1cm}} 1.41$$

Answer: _____

- Answer: _____

- Answer: _____